

# Does a mortality schedule need a Makeham term? Calibrating the likelihood-ratio test

Silvio C. Patricio

Interdisciplinary Center on Population Dynamics  
University of Southern Denmark

## Abstract

Whether a fitted mortality model needs a Makeham term, the non-negative constant representing background mortality, is commonly decided with a likelihood-ratio test. Because the constant cannot be negative, testing its absence places the parameter on the boundary of its range, and the usual chi-squared calibration does not apply. Assuming independent Poisson death counts and the stated regularity conditions, we show that when the constant is the only parameter on a boundary the statistic converges to an equal mixture of a point mass at zero and a chi-squared distribution with one degree of freedom, so that at the 5% level the critical value is 2.71 rather than the usual 3.84. Using 3.84 halves the asymptotic type I error rate and lowers the power to detect a positive Makeham term. We state the conditions under which the correction holds for Makeham models and verify the local information condition for Gompertz–Makeham and gamma-Gompertz–Makeham. We then quantify the information available about the constant, deriving its local power, an approximate detection threshold, and a bound on the information lost as the age window narrows. When a gamma-frailty variance is estimated and its true value is also zero, the limit is not generally a mixture of chi-squared distributions, and with negatively correlated adjusted scores the usual boundary correction rejects a true null hypothesis too often. Monte Carlo experiments show that correct calibration need not ensure adequate power at finite exposures, and that increasing exposure cannot compensate for using the wrong reference distribution.

**Keywords:** Makeham mortality models | Likelihood-ratio testing | Boundary parameters | Mortality heterogeneity | Statistical power

## 1 Introduction

In Makeham mortality models the force of mortality splits into an age-dependent component and a non-negative constant, and fitting one requires deciding whether the data support that constant. The decision changes what the rest of the model means. The Gompertz slope describes the exponential increase of the age-dependent component (Castellares et al., 2022), so omitting a constant that belongs leaves that component to absorb mortality attributable to background risk, and the fitted rate of aging moves with it.

Makeham (1860) introduced the constant by adding it to the exponentially increasing force of mortality of the Gompertz law. It is usually interpreted as background or extrinsic risk, the part of the force of mortality often associated with accidents and infections rather than with senescence, although those causes need not themselves have age-invariant rates. Patricio and Missov (2024) take the Makeham term as given and quantify

its contribution, through the share of deaths it accounts for at each age and the threshold age after which senescent mortality predominates. The question here is the prior one, whether the data support a positive Makeham term at all.

Setting the constant to zero reduces Gompertz–Makeham to Gompertz, so the two models are nested and the decision is commonly made with a likelihood-ratio test. Pletcher (1999) uses nested likelihood-ratio comparisons on age-specific mortality data to decide which components a model needs. Such comparisons support conclusions only if the reference distribution of the test statistic is correct.

We denote the Makeham parameter by  $\kappa$  and test

$$H_0 : \kappa = 0, \quad H_1 : \kappa > 0.$$

We restrict  $\kappa \geq 0$ , so  $H_0$  places it at the boundary of that range rather than in its interior. The usual result that the likelihood-ratio statistic has an asymptotic chi-squared distribution relies on the

restricted parameter lying in the interior, and it fails here (Chernoff, 1954; Self & Liang, 1987).

The boundary problem is familiar in mortality research, and it has been worked out most fully for the frailty variance of the gamma-Gompertz model, where testing for mortality deceleration means testing whether that variance is zero (Böhnstedt & Gampe, 2019; Böhnstedt et al., 2021). The same limit appears when a Gompertz hazard is tested against a constant hazard at extreme ages (Belzile et al., 2021). For the Makeham term, Daňko et al. (2018) halve the likelihood-ratio  $p$ -values in a gamma-Gompertz–Makeham model because the null lies on the boundary of the parameter space. This correction requires the frailty variance to be an interior nuisance parameter. If its true value is zero and we estimate it under both hypotheses, testing the Makeham term involves two boundary constraints.

When  $\kappa$  is the only parameter on a boundary and the regularity conditions in Appendix A hold, the test statistic converges to  $\frac{1}{2}\delta_0 + \frac{1}{2}\chi_1^2$  rather than to  $\chi_1^2$ , where  $\delta_0$  denotes a point mass at zero. Half of that limiting probability lies at zero, so for a positive observed statistic the  $\chi_1^2$  reference returns twice the  $p$ -value from the mixture, and a test calibrated that way at a nominal 5% level has an asymptotic type I error rate of 2.5%.

In practice the statistic should be compared with 2.71, the 90th percentile of  $\chi_1^2$ , rather than with 3.84, or equivalently the  $p$ -value from  $\chi_1^2$  should be halved when the statistic is positive. The conventional cutoff rejects less often under both hypotheses and misses some positive terms that the corrected test would detect. Every rejection it produces survives the correction; only some decisions to retain a Gompertz model can change.

The half-and-half limit for the Makeham term follows from the theory of Self and Liang (1987) once its conditions hold, and it is the starting point of this note rather than its main result. We make three further contributions. First, we state the conditions under which the limit holds for Poisson death counts and Makeham mortality models, and verify the local information condition for the Gompertz and gamma-Gompertz hazards, keeping global identification as a separate assumption. Second, we quantify the information available for detecting a Makeham term. Its information is the variation left after the age-dependent component has absorbed what it can, and it determines the local power of the test, an approximation to the smallest detectable Makeham term, and how

quickly detectability is lost as the age window narrows. Third, we show that when a gamma-frailty variance is estimated and its true value is zero, testing the Makeham term becomes a different nonregular problem. The limit is then not generally a mixture of chi-squared distributions, and the usual correction can make the test reject too often. Monte Carlo experiments illustrate the limits at finite exposures.

Section 2 develops the general likelihood-ratio test and applies it to Gompertz–Makeham and to gamma-Gompertz–Makeham with a positive frailty variance. Section 2.5 treats a frailty variance whose true value is zero, Section 2.6 the information available about the Makeham term, and Section 2.7 what the statistic responds to when the fitted model is wrong. Section 3 reports the numerical illustrations, and the appendices give the proofs.

## 2 Testing for a Makeham term

We observe death counts  $D_i$  in age intervals  $[x_i, x_i + 1)$ , for  $i = 1, \dots, m$ , and treat the exposure  $e_i > 0$  in the same interval as fixed and expressed in person-years. We write the population force of mortality as

$$\bar{\mu}(x; \kappa, \boldsymbol{\theta}) = \kappa + h(x, \boldsymbol{\theta}), \quad \kappa \geq 0, \quad (1)$$

where  $\kappa$  is the Makeham constant and  $h(x, \boldsymbol{\theta})$  is an age-dependent component, which may be Gompertz, gamma-Gompertz, the age-dependent part of Siler, or another mortality law. We write  $h_i(\boldsymbol{\theta}) = h(x_i, \boldsymbol{\theta})$  for the force of mortality at the fitted ages and treat  $\bar{\mu}(x_i; \kappa, \boldsymbol{\theta})$  as the rate applying throughout  $[x_i, x_i + 1)$ , the usual piecewise-constant approximation for single-year intervals. The vector  $\boldsymbol{\theta}$  represents the parameters of  $h$ , and where one of them is restricted to be positive, such as the Gompertz level  $\lambda$ , we take its logarithm, so that the true  $\boldsymbol{\theta}_0$  is interior to the fitted parameter set, as the conditions in Appendix A require.

Conditional on the exposures, we assume independent death counts with

$$D_i \sim \text{Poisson}(e_i \bar{\mu}(x_i; \kappa, \boldsymbol{\theta})),$$

which Brillinger (1986) motivates as an approximation for death counts given exposure. The likelihood is

$$L(\kappa, \boldsymbol{\theta}) = \prod_{i=1}^m \frac{\{e_i \bar{\mu}(x_i; \kappa, \boldsymbol{\theta})\}^{D_i}}{D_i!} e^{-e_i \bar{\mu}(x_i; \kappa, \boldsymbol{\theta})},$$

and, dropping factors free of the parameters, the log-likelihood is

$$\ell(\kappa, \boldsymbol{\theta}) = \sum_{i=1}^m [D_i \log\{\kappa + h_i(\boldsymbol{\theta})\} - e_i\{\kappa + h_i(\boldsymbol{\theta})\}]. \quad (2)$$

Maximum likelihood estimates follow from maximizing (2), and both hypotheses are fitted to the same data.

## 2.1 The statistic and its distribution under the null hypothesis

Let  $\hat{\boldsymbol{\theta}}_0$  be the maximizer of (2) when  $\kappa = 0$ , and let  $(\hat{\kappa}, \hat{\boldsymbol{\theta}})$  be the maximizer over  $\kappa \geq 0$  and  $\boldsymbol{\theta}$ . The statistic is

$$T = 2 \left\{ \ell(\hat{\kappa}, \hat{\boldsymbol{\theta}}) - \ell(0, \hat{\boldsymbol{\theta}}_0) \right\}, \quad (3)$$

which measures the improvement the Makeham parameter provides in fit to the death counts. The restricted model is a special case of the full one, so  $T$  is never negative, and larger values indicate stronger evidence for the Makeham term.

The test can only work if a Makeham term leaves a pattern in the data that the age-dependent component cannot produce on its own. Appendix A states this requirement formally, as the condition that the Fisher information matrix be positive definite.

To obtain the null distribution we reduce the maximization in (3) to one over  $\kappa$  alone. Holding  $\kappa \geq 0$  fixed and maximizing over the remaining parameters gives the profile log-likelihood

$$\ell_p(\kappa) = \max_{\boldsymbol{\theta}} \ell(\kappa, \boldsymbol{\theta}).$$

Thus  $T$  is twice the maximum of  $\ell_p(\kappa) - \ell_p(0)$  over  $\kappa \geq 0$ , which is why the quadratic approximation below is maximized over the same range. We define the profile score  $U$  for  $\kappa$  as the slope of the profile log-likelihood at  $\kappa = 0$ ,

$$U = \ell'_p(0),$$

and the observed profile information  $J$  as its negative curvature there,

$$J = -\ell''_p(0).$$

Both derivatives are taken from the right. A positive  $U$  means that a small increase in  $\kappa$  initially improves the fit, and both quantities already account for the estimation of  $\boldsymbol{\theta}$ .

The profile score is a weighted sum, over ages, of the differences between observed and expected

deaths, adjusted for having estimated  $\boldsymbol{\theta}$ . It is therefore approximately normal when exposures are large, and Appendix B makes this precise.

For the asymptotic argument we write  $e_{i,n} = n\bar{e}_i$ , where  $\bar{e}_i > 0$  are fixed baseline exposures and  $n$  is a dimensionless multiplier common to all ages, so that total person-years grow while the exposure proportions across ages stay fixed. We call this setting *proportional exposure growth*. For cohort data it corresponds to a larger birth cohort with the same survival pattern, and for period data to a larger population with the same age structure. Appendix B.4 shows that  $J$  is positive with probability tending to one and that

$$Z = \frac{U}{\sqrt{J}} \xrightarrow{d} \mathcal{N}(0, 1). \quad (4)$$

Under  $H_0$  the estimator  $\hat{\kappa}$  is of order  $n^{-1/2}$ . Setting  $\kappa = u/\sqrt{n}$  with  $u \geq 0$  fixed, a second-order Taylor expansion of the profile log-likelihood around  $\kappa = 0$  gives

$$\ell_p(\kappa) - \ell_p(0) = \kappa U - \frac{1}{2} \kappa^2 J + o_P(1), \quad (5)$$

where  $o_P(1)$  denotes a random remainder converging to zero in probability, uniformly over bounded  $u$  (see Van der Vaart (2000) for the underlying asymptotic theory).

Under  $H_0$  the model has no Makeham term, but random variation in the death counts will push the profile score above zero about half the time. When  $U \leq 0$  the quadratic (5) is maximized at  $\kappa = 0$ ; when  $U > 0$  it is maximized at  $\kappa = U/J$ . Doubling the maximum,

$$T = [Z]_+^2 + o_P(1), \quad (6)$$

where  $[z]_+ = \max(z, 0)$ . The maximized quadratic and profile gains differ by  $o_P(1)$ , so (6) holds without the two maximizers coinciding (Appendix B.2). A standard normal variable is negative with probability one half, producing a zero statistic in the limit, and conditional on being positive its square is chi-squared with one degree of freedom. Half of the null distribution therefore sits at a single point to which  $\chi_1^2$  assigns no probability.

**Theorem 2.1. (Makeham-term test).** *Assume conditions (A1)–(A4) in Appendix A, which require in particular that  $\boldsymbol{\theta}_0$  be interior to the fitted parameter set and that the Fisher information matrix be positive definite. Under  $H_0$  and proportional exposure growth,*

$$T \xrightarrow{d} \frac{1}{2} \delta_0 + \frac{1}{2} \chi_1^2, \quad (7)$$

where  $\delta_0$  denotes a point mass at zero.

The theorem restricts the functional form of  $h$  only through the conditions in Appendix A, so the correction is not specific to the Gompertz family and applies to any mortality law of the form (1). Appendices C.1 and C.3 verify the local information condition for the two models treated below; global identification remains a separate assumption. If a nuisance parameter itself lies at a bound, the theorem does not apply and a separate derivation is required; the case that arises in practice, a frailty variance whose true value is zero, is treated in Section 2.5. Appendix B gives the proof.

## 2.2 The hypothesis-testing rule

For a nominal level  $\alpha \in (0, \frac{1}{2})$ , we reject  $H_0$  when

$$T > \chi_{1, 1-2\alpha}^2, \quad (8)$$

where  $\chi_{1,q}^2$  is the  $q$  quantile of a chi-squared distribution with one degree of freedom. At  $\alpha = 0.05$  the critical value is 2.7055, compared with 3.8415 under the usual degrees-of-freedom rule. For an observed value  $t$  of the statistic, the asymptotic  $p$ -value is

$$p(t) = \begin{cases} \frac{1}{2}\mathbb{P}(V \geq t), & t > 0, \\ 1, & t = 0, \end{cases} \quad V \sim \chi_1^2. \quad (9)$$

The value at  $t = 0$  is the usual right-tailed  $p$ -value for a limiting distribution that places probability one half exactly at zero: a draw from that distribution is then at least as large as the observed statistic with probability one. For  $t > 0$  the rule amounts to halving the  $p$ -value obtained from  $\chi_1^2$ .

Rejection supports an age-independent background component in addition to the fitted age pattern. Non-rejection means the death counts do not provide sufficient evidence for that component over the fitted age range, and does not rule out an age-independent contribution to mortality. Both readings are conditional on the fitted age-dependent component, since the null hypothesis is that  $\kappa = 0$  within the chosen model and not that mortality has no age-independent part. Section 2.7 returns to this.

## 2.3 Gompertz–Makeham model

The Gompertz–Makeham model takes  $h(x, \theta) = \lambda e^{\beta x}$ , so that

$$\bar{\mu}(x; \lambda, \beta, \kappa) = \lambda e^{\beta x} + \kappa, \quad \lambda > 0, \beta > 0, \kappa \geq 0. \quad (10)$$

Under  $H_0$  the constant  $\kappa$  is zero and (10) reduces to the Gompertz model, while under  $H_1$  a positive Makeham term is present. Castellares et al. (2022) discuss the structural properties of this model. Substituting into (2) gives

$$\ell_{\text{GM}}(\lambda, \beta, \kappa) = \sum_{i=1}^m \left[ D_i \log\{\lambda e^{\beta x_i} + \kappa\} - e_i\{\lambda e^{\beta x_i} + \kappa\} \right],$$

and the statistic (3) compares the Gompertz–Makeham fit with the Gompertz fit obtained at  $\kappa = 0$ .

On the log scale the age-dependent component is the straight line  $\log h(x) = \log \lambda + \beta x$ , so  $\lambda$  fixes its height and  $\beta$  its slope. The question raised in Section 2.1 becomes whether shifting or tilting that line can mimic adding a constant to the force of mortality itself, which lifts the log hazard most at younger ages and has diminishing relative influence as age increases. Appendix C.1 shows that it cannot, once three distinct ages are observed with positive exposure.

**Proposition 2.2. (Gompertz–Makeham test).** *Assume that the true  $\lambda, \beta$  are strictly positive and that the identification and regularity conditions in Appendix A hold. Under  $H_0$  and proportional exposure growth,*

$$T \xrightarrow{d} \frac{1}{2}\delta_0 + \frac{1}{2}\chi_1^2.$$

## 2.4 Gamma-Gompertz–Makeham model

Gamma frailty bends the age pattern at the opposite end of the age range from the Makeham term. As individuals with higher frailty tend to die earlier within a cohort, the population hazard rises more slowly with age than the individual hazard, which produces deceleration and eventually a plateau (Vaupel et al., 1979). The Makeham constant instead lifts the log hazard at young adult ages. A test for  $\kappa$  with  $\gamma$  estimated therefore asks whether the data need an age-independent component on top of what selection already accounts for at the older ages.

We denote the gamma-frailty variance by  $\gamma \geq 0$  and define

$$g(x; \lambda, \beta, \gamma) = \frac{\lambda e^{\beta x}}{1 + \gamma \frac{\lambda}{\beta} (e^{\beta x} - 1)}. \quad (11)$$

Taking  $h(x, \theta) = g(x; \lambda, \beta, \gamma)$  in (1) gives the gamma-Gompertz–Makeham hazard

$$\bar{\mu}(x; \lambda, \beta, \gamma, \kappa) = \kappa + g(x; \lambda, \beta, \gamma),$$

in which frailty acts on the Gompertz component while the Makeham term remains additive. For  $\gamma > 0$ ,  $g$  tends to  $\beta/\gamma$  as age increases, so the constant here raises the plateau from  $\beta/\gamma$  to  $\kappa + \beta/\gamma$  instead of becoming negligible at the oldest ages, which is how this case differs from the Gompertz one.

This selection argument follows a cohort as it ages. A period schedule combines different cohorts, so a gamma-Gompertz fit to period rates describes their age pattern without, by itself, identifying frailty variation within a cohort.

We test  $H_0 : \kappa = 0$  against  $H_1 : \kappa > 0$ , estimating  $\gamma$  as a nuisance parameter. A positive true variance is treated here, and a zero one in Section 2.5.

If the true frailty variance  $\gamma_0$  is strictly positive,  $\kappa$  remains the only parameter on a boundary, since  $\gamma$  can be perturbed in both directions and so behaves like an ordinary nuisance parameter. Theorem 2.1 then applies with  $\theta = (\log \lambda, \beta, \gamma)^\top$ .

**Proposition 2.3. (Gamma-Gompertz–Makeham test with positive frailty variance).** *Under  $H_0$ , assume  $\gamma_0 > 0$  and that the identification and Fisher information conditions in Appendix A hold for the nuisance vector  $(\log \lambda, \beta, \gamma)^\top$ . Under proportional exposure growth,*

$$T \xrightarrow{d} \frac{1}{2}\delta_0 + \frac{1}{2}\chi_1^2.$$

Here the information condition asks whether changes in  $\lambda, \beta, \gamma$  can reproduce a constant addition at every age. Appendix C.3 gives the derivatives and shows that four distinct ages with positive exposures are enough, provided  $\lambda\gamma \neq \beta$ .

## 2.5 When the frailty variance is also on the boundary

A frailty variance of zero is a realistic possibility whenever the data show little deceleration, and whether frailty is needed is itself the subject of dedicated tests (Böhnhstedt & Gampe, 2019; Patrio & Missov, 2023). If  $\gamma$  is fixed at zero in both fits, the test reduces to the Gompertz–Makeham case of Section 2.3. If instead  $\gamma \geq 0$  is estimated and its true value is zero, both  $\kappa$  and  $\gamma$  lie on their boundaries under  $H_0$ , and the distribution of (3) must account for both constraints.

Tests of a variance component often run into the issue that the remaining parameters become unidentified when the variance is zero, which invalidates the usual expansions. That does not

happen here. At  $\gamma = 0$ , (11) reduces to the Gompertz hazard, whose parameters  $\lambda$  and  $\beta$  remain identified, and a small positive  $\gamma$  changes the age pattern smoothly, so the usual expansions still apply. The problem is therefore a boundary problem of the Self and Liang (1987) type rather than a non-identifiability problem.

Appendix D derives the limiting statistic  $T_0$  for the case  $\kappa = \gamma = 0$ . Its distribution depends on  $\mathbf{K}$ , the efficient Fisher information matrix for  $(\kappa, \gamma)^\top$ . Writing  $F_{\mathbf{K}}$  for the distribution function of  $T_0$ , the rejection rule at a true frailty variance of zero is

$$T > F_{\mathbf{K}}^{-1}(1 - \alpha). \quad (12)$$

The distribution of  $T_0$  is not in general a mixture of chi-squared distributions. Under  $H_0$  we still estimate  $\gamma \geq 0$ , so the null parameter set is an edge of the quadrant rather than its origin, and that is what breaks the mixture form. When both parameters are fixed at zero under the null hypothesis, the limit is still such a mixture, with weights that depend on the information matrix (Self & Liang, 1987; Silvapulle & Sen, 2005). In particular, the critical value 2.7055 from Section 2.2 does not carry over. For a general  $\mathbf{K}$ ,  $F_{\mathbf{K}}$  should be evaluated by simulation from the explicit expression for  $T_0$  derived in Appendix D.2, and Appendix D.3 treats the special case in which the earlier mixture is recovered.

For the Gompertz hazard, changes in the Makeham term and frailty can partly compensate for one another after we fit the level and slope. A small  $\kappa$  bends the log hazard upward at young ages, making it convex, whereas a small  $\gamma$  bends it downward at old ages, making it concave. These opposite curvatures suggest a negative covariance  $k_{12}$  between the two adjusted scores, which Section 3.3 confirms for its simulation design. When  $k_{12} < 0$ , Appendix D.3 shows that the limiting law places probability  $\frac{1}{2} + \arcsin(\rho)/(2\pi) < \frac{1}{2}$  at zero, where  $\rho = k_{12}/\sqrt{k_{11}k_{22}}$  is the correlation of the two adjusted scores and the arcsine term comes from Sheppard’s classical formula for the orthant probabilities of a bivariate normal distribution (Sheppard, 1899), and that its upper tail lies above that of  $\frac{1}{2}\delta_0 + \frac{1}{2}\chi_1^2$  at every point. The rule (8) then rejects a true null hypothesis more often than its nominal level, at every level  $\alpha$ .

These results distinguish a true frailty variance of zero from a fixed positive one, but a fitted value cannot. A small  $\hat{\gamma}$  is compatible with a positive true variance, where Theorem 2.1 applies, and with a zero one, where it does not, so choosing

the critical value from  $\hat{\gamma}$  alone has no established guarantee of keeping the false-detection probability at  $\alpha$ . Appendix D.4 shows why variances approaching zero require further analysis.

## 2.6 Where the evidence about $\kappa$ comes from

A Makeham term raises mortality by the same absolute amount at every age. On the log scale this addition is large relative to the age-dependent hazard at young adult ages and shrinks as that hazard grows. Its share of the force of mortality at age  $x$ ,  $\kappa/\{\kappa + h(x, \boldsymbol{\theta})\}$ , is the expected proportion of deaths at that age that the model attributes to the age-independent component (Patricio & Missov, 2024). Under a Gompertz hazard the share declines to zero at the oldest ages. Under gamma frailty it tends instead to  $\kappa/(\kappa + \beta/\gamma)$ , because the constant raises the plateau (Section 2.4). The information available about  $\kappa$  depends also on the exposures and on the other parameters of  $h$ .

### 2.6.1 Information after estimating the age-dependent component

Appendix A writes the efficient information for  $\kappa$  as a weighted least-squares quantity in which age  $i$  carries the weight  $\bar{e}_i/h_i$ . What the weights multiply, however, is the residual left after projecting the constant vector on the nuisance derivative directions, and that residual depends on  $h$ .

For the Gompertz component of Section 2.3,  $\mathbf{r}_i = (h_i, x_i h_i)^\top$ , so the residual is  $1 - h_i(q_1 + q_2 x_i)$  at the minimizing coefficients  $q_1, q_2$ . These coefficients depend on the mortality schedule and exposures across all fitted ages, so a small  $h_i$  does not by itself make the residual close to one, and the weights determine how much each age contributes.

For a fixed age range and exposure pattern, total expected efficient information at the null is  $n j_{\text{eff}}$ . Whenever  $j_{\text{eff}} > 0$ , increasing exposure improves precision and the ability to detect small Makeham terms. Comparing age windows requires evaluating both the weights and the residuals in (A.4); the formula alone does not rank young and old ages by information per person-year.

### 2.6.2 Local power and the detection threshold

The same quantity determines the size of Makeham term the test can detect. Böhnhstedt and Gampe (2019) derive the local power of the boundary likelihood-ratio test for the frailty variance of

the gamma-Gompertz model, and the same argument applies to  $\kappa$  (Appendix B.3). For a term of order  $n^{-1/2}$ ,  $\kappa = u/\sqrt{n}$ , the rule (8) rejects with probability tending to  $\Phi(u\sqrt{j_{\text{eff}}} - z_{1-\alpha})$ , where  $\Phi$  is the standard normal distribution function and  $z_{1-\alpha}$  its  $1 - \alpha$  quantile, so that  $z_{0.95}^2 = 2.7055$ . Setting this probability equal to a target power  $\pi$  and solving for  $\kappa$  gives a local approximation to the smallest Makeham term detected with that power,

$$\kappa_{\min} \approx \frac{z_{1-\alpha} + z_\pi}{\sqrt{n j_{\text{eff}}}},$$

where  $z_\pi$  is the  $\pi$  quantile of the standard normal distribution. With  $\alpha = 0.05$  and the conventional target  $\pi = 0.8$  (Cohen, 1988),  $z_{0.80} = 0.84$  and  $\kappa_{\min} \approx 2.486/\sqrt{n j_{\text{eff}}}$ . The conventional critical value 3.8415 raises the threshold and so lowers power (Böhnhstedt et al., 2021). It replaces  $z_{0.95}$  by  $z_{0.975} = 1.96$ , and for a term of size  $\kappa_{\min}$  the local-power approximation gives a rejection probability of about 0.70 rather than 0.80.

### 2.6.3 The age window

Power also depends on the age window. Böhnhstedt et al. (2021) use the Fisher information to show how the power to detect mortality deceleration changes with the age range of the data, and the same reasoning applies to  $\kappa$  through  $j_{\text{eff}}$ . For the Gompertz hazard, narrowing the window has a strong effect. Over a short range of ages, a small change in the Gompertz level and slope reproduces a constant addition almost exactly, since near any age  $x_0$  the combination  $\{1 - \beta(x - x_0)\}e^{\beta(x - x_0)}$  of  $h$  and  $xh$  equals  $1 - O\{(x - x_0)^2\}$  (Appendix C.2). Little is then left for  $\kappa$  to explain,  $j_{\text{eff}}$  is small, and a non-rejection over a short window can reflect low power rather than a negligible Makeham term.

## 2.7 What the statistic responds to

The null hypothesis of Theorem 2.1 is that  $\kappa = 0$  within the chosen model, and the statistic compares two fits of that model. It therefore responds to any systematic difference between the data and  $h$  that a constant can partly absorb, not only to an age-independent component of mortality. A rejection or a non-rejection is conditional on  $h$  describing the age pattern over the fitted window.

Departures at the young end of the window can move the statistic in either direction. Childhood mortality or an accident hump raises mortality above a Gompertz line at some young ages and not at others, so its effect on  $\hat{\kappa}$  depends on the window

and on the shape of the excess. Fitting Gompertz–Makeham from age 20, Patricio and Missov (2024) found the Makeham term underestimated rather than inflated in populations with an irregular age pattern between ages 20 and 45.

Old-age deceleration can conceal a Makeham contribution when we fit Gompertz–Makeham. The log hazard of Gompertz–Makeham is convex in age, since  $\kappa$  lifts it most where the Gompertz component is small, whereas deceleration makes the observed log hazard concave at old ages. A positive  $\kappa$  cannot reproduce that concavity. When deceleration dominates the fitted window, the constrained fit can place  $\hat{\kappa}$  at zero and give  $T = 0$ , even when an age-independent component is present. The outcome depends on the age window, exposures and mortality schedule. When the fitted window reaches ages with deceleration, the Makeham term should be tested within a model that allows for it, such as gamma-Gompertz–Makeham.

### 3 Numerical illustrations

We use Monte Carlo experiments to illustrate the consequences of the reference distribution and available exposure for detecting a Makeham term. The simulations address three points: false detections when the Makeham term is the only parameter on a boundary, the power to detect a positive term, and the change in calibration when we also estimate a frailty variance whose true value is zero. The aim is illustration rather than validation of the asymptotic results established in Section 2.

#### 3.1 Finite-sample calibration

We first simulate death counts under a Gompertz hazard with no Makeham term,

$$D_{i,n} \sim \text{Poisson} \{n\bar{e}_i\lambda_0 \exp(\beta_0 x_i)\},$$

for single years of age  $x_i \in [30, 90]$ , with  $\lambda_0 = 2 \times 10^{-5}$  and  $\beta_0 = 0.10$ . We take the baseline exposures proportional to the survivorship implied by the same schedule,  $\bar{e}_i \propto \exp\{-(\lambda_0/\beta_0)(e^{\beta_0 x_i} - e^{30\beta_0})\}$ , and normalize them to sum to one, so that  $n$  is the total exposure in person-years and the design introduces no parameter beyond the schedule itself. We vary  $n \in \{10^4, 10^5, 10^6, 10^7\}$  while keeping these exposure proportions fixed, as in the asymptotic framework of Section 2. Throughout the numerical illustrations, an increase in exposure means an increase in this total  $n$ , not a change in its distribution across ages. For each value of  $n$  we

generate  $B = 10,000$  independent samples and fit the Gompertz and Gompertz–Makeham models.

**Table 1:** Finite-sample calibration of the Makeham likelihood-ratio test.

| $n$        | $\Pr(T \approx 0)$ | $\Pr(T > 2.7055)$ | $\Pr(T > 3.8415)$ | Failed |
|------------|--------------------|-------------------|-------------------|--------|
| $10^4$     | 0.566              | 0.041             | 0.021             | 0      |
| $10^5$     | 0.518              | 0.043             | 0.024             | 1      |
| $10^6$     | 0.511              | 0.052             | 0.027             | 3      |
| $10^7$     | 0.496              | 0.049             | 0.023             | 8      |
| Asymptotic | 0.500              | 0.050             | 0.025             | –      |

$B = 10,000$  per exposure level. Probabilities use successful fits; failures are shown separately. We classify  $|T_{\text{raw}}| \leq 1 \times 10^{-7}$  as numerical zero.

Table 1 compares the empirical rejection probabilities under the boundary-corrected cutoff 2.7055 and the conventional  $\chi_1^2$  cutoff 3.8415. The corrected test rejects slightly less often than the nominal 5% level at the two smaller exposures, with rejection probabilities of 0.041 and 0.043. At the two larger exposures, these probabilities are 0.052 and 0.049, close to the nominal level. Using the conventional cutoff instead gives rejection probabilities between 0.021 and 0.027, consistent with its asymptotic rejection probability of 0.025.

A small number of fits failed to converge. The count rises with exposure, which is consistent with a sharper likelihood and fixed convergence tolerances, and at no more than eight failures in 10,000 samples it does not affect the reported probabilities. We classify a statistic as numerical zero when  $|T_{\text{raw}}| \leq 10^{-7}$ , where  $T_{\text{raw}}$  is twice the fitted log-likelihood difference before numerical truncation at zero. The proportion of such statistics approaches one half as  $n$  increases, consistent with the limiting law

$$\frac{1}{2}\delta_0 + \frac{1}{2}\chi_1^2.$$

For this design, the boundary correction provides approximately nominal calibration at the larger exposures, whereas the conventional cutoff remains conservative.

Increasing exposure therefore improves the accuracy of the boundary approximation in this design, but does not make the conventional cutoff appropriate for a 5% test. The smaller-exposure results also show why we should distinguish the nominal level from the actual probability of a false detection. Adequate control of false detections does not, however, tell us whether a positive component is large enough to be detected.

### 3.2 Local power and detectability

We next measure power: the probability that the test detects a Makeham term when a positive term is present. We generate counts under local alternatives

$$\kappa_n = \frac{u}{\sqrt{n}},$$

with  $u > 0$ , so the positive constant approaches zero as total exposure  $n$  grows. We evaluate the efficient information  $j_{\text{eff}}$  under the null using (A.4) and express the alternatives through the standardized shift

$$\delta = u\sqrt{j_{\text{eff}}},$$

which combines the size of the Makeham term with the information available to detect it after estimating the age-dependent component. The local-power approximation is

$$\pi(\delta) = \Phi(\delta - z_{1-\alpha}).$$

At a fixed  $\delta$ , the Makeham constant decreases as  $n$  increases. We therefore assess the detection of progressively smaller age-independent contributions to mortality, rather than hold the mortality schedule fixed across exposure levels.

Figure 1 shows that empirical power approaches the local approximation as  $n$  increases. For this design, the approximation overstates power, particularly at the smallest exposure. Part of that gap is a difference in level rather than in power, since at  $n = 10^4$  the corrected test rejects in 4.1% of null samples rather than 5% (Table 1), so the same samples are judged against an effectively stricter test than the approximation assumes.

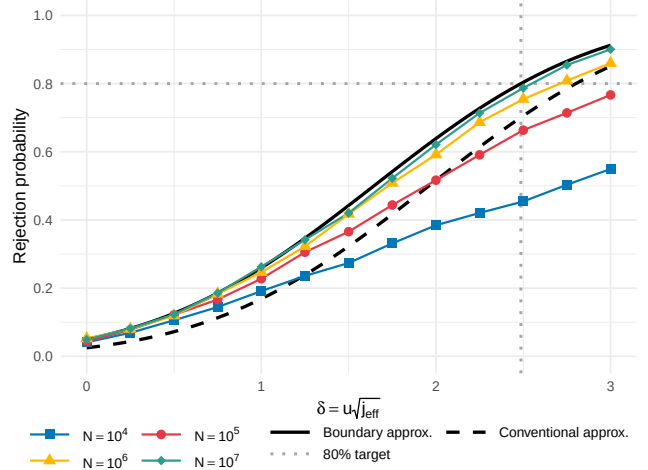
At  $\alpha = 0.05$  and target power  $\pi = 0.80$ , Section 2.6 gives the approximate detection threshold

$$\kappa_{\min} \approx \frac{z_{0.95} + z_{0.80}}{\sqrt{n j_{\text{eff}}}}.$$

For this design,  $j_{\text{eff}} \approx 243.96$ . At the corresponding thresholds, empirical power using the corrected cutoff increases from approximately 46% at  $n = 10^4$  to 78% at  $n = 10^7$ , approaching the target of 80%. Applying the conventional cutoff to the same simulated samples lowers power by approximately 11 percentage points across all four exposure totals.

These results show why failure to detect a Makeham term need not imply that its contribution to mortality is negligible. At the smallest exposure, the corrected test fails to detect the positive term in more than half the simulated samples, even though the local approximation predicts

Local power of the Makeham likelihood-ratio test



**Figure 1:** Local power of the Makeham likelihood-ratio test. Coloured curves show empirical rejection probabilities using the boundary-corrected cutoff 2.7055 at each exposure level. The solid and dashed black curves show the local-power approximations for the corrected and conventional cutoffs, respectively. The dotted lines mark the target power of 0.80 and its corresponding standardized shift,  $\delta \approx 2.486$ .

80% power. A decision to retain the Gompertz model can therefore reflect limited information rather than the absence of an age-independent component.

When comparing mortality schedules across populations or periods with different exposures, we must also account for differences in detectability. A Makeham term may be supported in one dataset and remain undetected in another because the available information differs. Such differences in test outcomes do not, by themselves, establish differences in the underlying mortality components.

The boundary correction reduces missed detections, but it does not remove this dependence on the available information. We therefore interpret a non-rejection alongside the size of the contribution the data could plausibly detect, using the analytical threshold as an approximate guide whose accuracy improves with exposure in this design.

### 3.3 The joint boundary at zero frailty

Finally, we consider the setting of Section 2.5, in which both the Makeham constant and the true frailty variance are zero. We generate counts from the same Gompertz schedule with

$$\kappa_0 = \gamma_0 = 0.$$

This generating point belongs to the null hypothesis  $\kappa = 0$ , which leaves the frailty variance

unknown. Under the null we therefore fit gamma-Gompertz with  $\kappa = 0$  and estimate  $\gamma \geq 0$ . Under the alternative, we fit gamma-Gompertz–Makeham and estimate both  $\kappa \geq 0$  and  $\gamma \geq 0$ . We test for the Makeham term alone, while allowing for frailty under both hypotheses. Fixing  $\gamma = 0$  in both fits would instead recover the Gompertz–Makeham test of Section 3.1.

For this design, the efficient information matrix  $\mathbf{K}$  gives the adjusted-score correlation

$$\rho = \frac{k_{12}}{\sqrt{k_{11}k_{22}}} \approx -0.630.$$

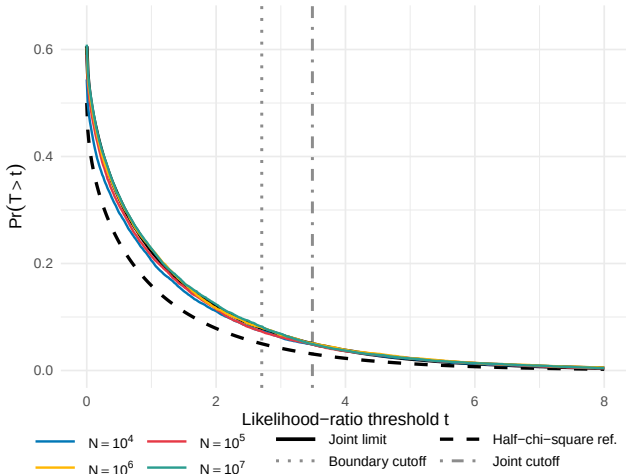
The negative correlation reflects the interaction between the two components after accounting for the Gompertz level and slope. As Appendix D shows, this interaction changes the null distribution of the Makeham test.

We compare the finite-sample statistic with the joint-boundary limit by drawing 100,000 realizations of

$$\mathbf{W} \sim \mathcal{N}_2(\mathbf{0}, \mathbf{K})$$

and evaluating  $T_0(\mathbf{W})$ . Figure 2 shows the corresponding upper-tail probabilities. At any threshold  $t$ , the height of a curve gives the probability of rejecting the null when we use  $t$  as the critical value.

**Makeham test with zero true frailty variance**



**Figure 2:** Null distribution of the Makeham likelihood-ratio statistic when the true frailty variance is zero. Coloured curves show finite-sample upper-tail probabilities at each exposure level. The solid black curve shows the simulated joint-boundary limit, and the dashed black curve shows the half-and-half chi-squared reference distribution. The vertical lines mark the usual boundary cutoff 2.7055 and the simulated joint-boundary cutoff 3.489. Frailty remains estimated under both hypotheses.

The finite-sample curves approach the joint-boundary limit as exposure increases. The proportion of statistics classified as numerical zero de-

creases from 0.456 at  $n = 10^4$  to 0.391 at  $n = 10^7$ , close to the theoretical value

$$\Pr(T_0 = 0) = \frac{1}{2} + \frac{\arcsin(\rho)}{2\pi} \approx 0.391.$$

The usual half-and-half law instead assigns probability 0.5 to zero and understates the upper-tail probabilities for this design.

This difference affects the frequency with which we infer a Makeham component. The cutoff 2.7055 leaves 5% of the half-and-half reference distribution above it, but approximately 7.9% of the joint-boundary limit. With this cutoff, the empirical rejection probability should therefore approach 0.079, not 0.05, as exposure increases. The simulated rejection probabilities, between 0.073 and 0.082 across the four exposure totals, follow this pattern. The test detects a Makeham term in roughly eight out of every hundred samples, although the generating schedule contains none and the nominal level allows five. Increasing exposure cannot remove this excess because it arises from the wrong limiting distribution.

The simulated 95th percentile of the joint-boundary law is 3.489. Using this cutoff gives empirical rejection probabilities between 0.049 and 0.051, close to the nominal level. Here we calculate  $\mathbf{K}$  from the known generating parameters, so this comparison isolates the effect of the reference distribution. It does not establish the performance of a procedure that estimates  $\mathbf{K}$  or selects a calibration from the fitted frailty variance.

Allowing for frailty therefore changes more than the shape of the fitted mortality schedule: it can also change the evidence required to support an additional age-independent component. When the true frailty variance is zero but we estimate it, the usual Makeham boundary correction can mistake random variation for evidence of that component too often. These results distinguish this setting from one in which we fix frailty at zero before fitting, and they do not justify choosing between reference distributions solely according to whether the fitted variance equals zero.

## 4 Discussion

When the Makeham constant is the only parameter on a boundary, the likelihood-ratio statistic for its absence converges to  $\frac{1}{2}\delta_0 + \frac{1}{2}\chi_1^2$ . The practical rule follows directly. At the 5% level the statistic should be compared with 2.71, or the  $p$ -value from  $\chi_1^2$  halved when the statistic is positive, while using the 95th percentile of  $\chi_1^2$ , 3.84, lowers both the

asymptotic type I error rate and the power of the test. The result holds for Makeham models of the form  $\kappa + h(x, \theta)$  under the stated conditions, and we have verified its local information condition for Gompertz–Makeham and gamma-Gompertz–Makeham.

The calibration is only part of what determines a conclusion. The information about  $\kappa$  is the variation left after the age-dependent component has absorbed what it can, so the same test can be decisive on a wide age window and nearly powerless on a narrow one, and a non-rejection has to be read against the approximate size of the Makeham term the test could detect at a specified power. A correctly calibrated test can still be badly underpowered. At the smallest exposure considered here it misses a term at its own nominal detection threshold in more than half of the samples, so retaining a Gompertz model may record the limits of the data rather than the absence of a background component.

Two further conditions govern what a rejection means. The test is conditional on the fitted model, since old-age deceleration fitted with Gompertz–Makeham can push the estimate of  $\kappa$  to zero. And when a gamma-frailty variance is estimated at a true value of zero, the half-and-half correction is not generally valid and rejects too often whenever the two adjusted scores are negatively correlated. That excess does not diminish as exposure grows, because it comes from using the wrong limiting distribution rather than from a finite-sample approximation.

Deciding whether a mortality schedule needs a Makeham term therefore requires the correct reference distribution, an age window wide enough to detect the term, and a model for the age-dependent component that matches the data. What follows from the decision reaches beyond the constant itself. When the term is dropped, the Gompertz component takes up what it would have carried, and the fitted rate of aging moves with it, so the calibration of this test bears on the quantity that mortality models are most often fitted to estimate.

The results are asymptotic, under proportional growth of exposures across ages, and the two frailty cases treated here leave out a variance that is unknown and arbitrarily close to zero, for which controlling false detections requires an argument holding uniformly across such sequences. We have also assumed independent Poisson death counts, the standard sampling model in mortality estimation, although real counts are sometimes overdispersed

relative to Poisson. The half-and-half limit comes from the constraint  $\kappa \geq 0$  rather than from the Poisson assumption, so the boundary argument itself would carry over to another count model, but the remaining steps of the proof would have to be established again.

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**Data and materials availability:** The full analysis is openly available at

[https://github.com/scpatricio/Makeham\\_LRT](https://github.com/scpatricio/Makeham_LRT)

so that results can be reproduced.

**Author contributions:** Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Writing – original draft, Writing – review and editing, and Visualization: S.C.P.

## Proofs and technical results

### A Assumptions and Fisher information

This appendix sets up the asymptotic framework, states the regularity conditions (A1)–(A4) invoked by Theorem 2.1, and defines the Fisher information matrix together with the efficient information  $j_{\text{eff}}$  for  $\kappa$ , in the weighted least-squares form used in Section 2.6.

We keep the  $m$  age intervals fixed and increase their exposures proportionally. For fixed baseline exposures  $\bar{e}_i > 0$ , we define

$$e_{i,n} = n\bar{e}_i. \quad (\text{A.1})$$

We write  $\boldsymbol{\theta} \in \mathbb{R}^p$  for the vector of  $p$  parameters of the age-dependent component and  $h_i(\boldsymbol{\theta}) = h(x_i, \boldsymbol{\theta})$ . Conditional on these exposures, we assume independent random death counts with

$$D_{i,n} \sim \text{Poisson}(n\bar{e}_i\{\kappa + h_i(\boldsymbol{\theta})\}).$$

Every likelihood in the note follows by specifying  $h_i$  in this model. We write  $\bar{\mu}_i(\kappa, \boldsymbol{\theta}) = \kappa + h_i(\boldsymbol{\theta})$  for the population force of mortality at age  $x_i$  and take  $n \rightarrow \infty$ . Total exposure equals  $n \sum_i \bar{e}_i$ , so  $n$  scales total person-years without changing their age distribution.

We use the following sufficient conditions for Theorem 2.1:

- (A1) We fit the model over  $[0, \bar{\kappa}] \times \mathcal{T}$ , where  $\bar{\kappa} > 0$  and  $\mathcal{T} \subset \mathbb{R}^p$  is compact. The set  $\mathcal{T}$  contains an open ball around the true  $\boldsymbol{\theta}_0$ . Thus  $\boldsymbol{\theta}$  can be varied in every direction by a sufficiently small amount, while the constraint  $\kappa \geq 0$  remains binding at  $\kappa = 0$ .
- (A2) At every fitted age,  $\bar{\mu}_i(\kappa, \boldsymbol{\theta})$  depends continuously on the parameters, so small parameter changes produce small changes in the force of mortality. Two distinct parameter vectors cannot give the same values at all fitted ages. There is also a constant  $\varepsilon > 0$  such that  $\bar{\mu}_i(\kappa, \boldsymbol{\theta}) \geq \varepsilon$  for every fitted age and every parameter vector in the fitted set.
- (A3) Each  $h_i(\boldsymbol{\theta})$  has continuous partial derivatives with respect to  $\boldsymbol{\theta}$  up to order three near  $\boldsymbol{\theta}_0$ . The normalized Fisher information matrix below is positive definite.
- (A4) Under each hypothesis, we use a parameter vector that gives the largest log-likelihood over its entire fitted parameter set. A numerical approximation is sufficient if the difference between that largest value and the value returned by the optimizer converges to zero in probability, that is, is  $o_P(1)$ .

Compactness is used to control the maximizers; suitable consistency conditions can replace it on an unbounded parameter set.

Under  $H_0$ , we write  $h_i = h_i(\boldsymbol{\theta}_0)$  and define the fixed derivative vector

$$\mathbf{r}_i = \left. \frac{\partial h_i(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}} \right|_{\boldsymbol{\theta}=\boldsymbol{\theta}_0}.$$

Differentiating (2) gives the scalar score for  $\kappa$  and

the vector score for  $\boldsymbol{\theta}$ ,

$$U_\kappa = \sum_i \left( \frac{D_{i,n}}{h_i} - n\bar{e}_i \right),$$

$$\mathbf{U}_\theta = \sum_i \left( \frac{D_{i,n}}{h_i} - n\bar{e}_i \right) \mathbf{r}_i,$$

both of which are weighted sums of the centred residuals  $D_{i,n} - n\bar{e}_i h_i$ , which yields the normality established in Appendix B.

We define Fisher information as the expected negative Hessian of the log-likelihood; it measures the expected curvature around the true parameter (Van der Vaart, 2000, Chapter 7). Dividing by  $n$  gives the fixed matrix

$$\mathbf{I} = \sum_i \frac{\bar{e}_i}{h_i} \begin{pmatrix} 1 \\ \mathbf{r}_i \end{pmatrix} \begin{pmatrix} 1 \\ \mathbf{r}_i \end{pmatrix}^\top. \quad (\text{A.2})$$

The entry  $i_{\kappa\kappa}$  measures information about  $\kappa$  when  $\boldsymbol{\theta}$  is known. When we estimate  $\boldsymbol{\theta}$ , some of the variation in the score for  $\kappa$  can also be explained by the nuisance scores. We subtract that linear prediction and use the variance of the remaining score, divided by  $n$ . Partitioning  $\mathbf{I}$  according to  $(\kappa, \boldsymbol{\theta}^\top)^\top$ , this gives the normalized efficient information

$$j_{\text{eff}} = i_{\kappa\kappa} - \mathbf{i}_{\theta\kappa}^\top \mathbf{I}_{\theta\theta}^{-1} \mathbf{i}_{\theta\kappa}. \quad (\text{A.3})$$

The subtracted term is the information lost through estimating  $\boldsymbol{\theta}$ ; the resulting matrix expression is the Schur complement. The same quantity has a weighted least-squares form:

$$j_{\text{eff}} = \min_{\mathbf{q} \in \mathbb{R}^p} \sum_i \frac{\bar{e}_i}{h_i} \{1 - \mathbf{r}_i^\top \mathbf{q}\}^2. \quad (\text{A.4})$$

Expanding the squared residuals in (A.4) gives

$$i_{\kappa\kappa} - 2\mathbf{q}^\top \mathbf{i}_{\theta\kappa} + \mathbf{q}^\top \mathbf{I}_{\theta\theta} \mathbf{q}.$$

Its minimizing coefficient vector is  $\mathbf{I}_{\theta\theta}^{-1} \mathbf{i}_{\theta\kappa}$ . Substituting this vector gives (A.3).

The constant vector represents an equal increase in the force of mortality at every age. The derivative columns represent the changes produced by the nuisance parameters. Equation (A.4) measures the weighted difference left after approximating the first with the second. If the nuisance information matrix is nonsingular,  $j_{\text{eff}} > 0$  exactly when the derivative columns cannot reproduce the constant vector. This local requirement is separate from the global identification condition in (A2).

## B Proof of Theorem 2.1

We establish score normality, consistency and the local quadratic expansion, then maximize the quadratic over  $u \geq 0$ . This retains only the positive part of the efficient score and produces the point mass at zero. The remaining subsections derive local power and connect the argument to the profile quantities  $U$  and  $J$  of Section 2.1.

### B.1 Score, consistency and local expansion

We collect the parameters in  $\zeta = (\kappa, \theta^\top)^\top$  and denote their true values under  $H_0$  by  $\zeta_0$ . We define the fixed vector

$$\mathbf{b}_i = \begin{pmatrix} 1 \\ \mathbf{r}_i \end{pmatrix}.$$

The normalized random score vector is

$$\mathbf{S}_n = \sum_i \frac{D_{i,n} - n\bar{e}_i h_i}{\sqrt{n} h_i} \mathbf{b}_i.$$

Each count has a mean tending to infinity; centring and standardizing gives a normal limit. Independence across ages and the multivariate convergence argument (Van der Vaart, 2000, Chapter 2) then give

$$\mathbf{S}_n \xrightarrow{d} \mathcal{N}_{p+1}(\mathbf{0}, \mathbf{I}). \quad (\text{B.1})$$

For consistency, we first use  $D_{i,n}/n \xrightarrow{P} \bar{e}_i h_i$ , the fixed number of ages, and the bounds in (A2). Together with continuity and compactness, these give uniform convergence of  $n^{-1}\{\ell_n(\zeta) - \ell_n(\zeta_0)\}$  to

$$\ell_\infty(\zeta) = \sum_i \bar{e}_i \left[ h_i \log \left\{ \frac{\bar{\mu}_i(\zeta)}{h_i} \right\} - \{\bar{\mu}_i(\zeta) - h_i\} \right].$$

Each summand is nonpositive and equals zero only when  $\bar{\mu}_i(\zeta) = h_i$ . Identification makes  $\zeta_0$  the unique maximizer. Uniform convergence and (A4) therefore give consistency of both estimators by the argmax consistency argument (Van der Vaart, 2000, Chapter 5).

At the true parameter,  $\mathbf{I}$  is the probability limit of the negative Hessian divided by  $n$ . Terms involving second derivatives multiply centred count residuals and vanish after normalization. The derivative continuity in (A3) gives the corresponding local convergence. Positive definiteness means that the expected log-likelihood decreases quadratically in every parameter direction away from the true value. Its curvature grows at rate  $n$ , whereas

the score is  $O_P(\sqrt{n})$ . Together with consistency, these orders imply that the estimation errors are  $O_P(n^{-1/2})$ ; the boundary restriction changes the admissible directions, not this rate (Self & Liang, 1987).

We express parameter displacements on this scale by defining

$$u = \sqrt{n} \kappa,$$

$$\mathbf{t} = \sqrt{n}(\theta - \theta_0).$$

We hold  $u \geq 0$  and  $\mathbf{t} \in \mathbb{R}^p$  fixed as  $n$  increases, so the corresponding parameters approach the null at rate  $n^{-1/2}$ . Writing  $S_{\kappa,n}$  and  $\mathbf{S}_{\theta,n}$  for the corresponding blocks of  $\mathbf{S}_n$ , Taylor expansion gives, uniformly for bounded displacements,

$$\begin{aligned} \ell_n\left(\frac{u}{\sqrt{n}}, \theta_0 + \frac{\mathbf{t}}{\sqrt{n}}\right) - \ell_n(0, \theta_0) \\ = u S_{\kappa,n} + \mathbf{t}^\top \mathbf{S}_{\theta,n} \\ - \frac{1}{2} \left( u^2 i_{\kappa\kappa} + 2u \mathbf{i}_{\theta\kappa}^\top \mathbf{t} + \mathbf{t}^\top \mathbf{I}_{\theta\theta} \mathbf{t} \right) + o_P(1). \end{aligned} \quad (\text{B.2})$$

Bounds on third derivatives control the remainder. The  $O_P(n^{-1/2})$  estimation errors allow us to evaluate this expansion at the estimators, whose scaled displacements remain bounded in probability.

### B.2 Maximization under the boundary constraint

Let  $Q_n(u, \mathbf{t})$  denote the quadratic expression on the right-hand side of (B.2), excluding the remainder  $o_P(1)$ . For fixed  $u$ , maximizing this quadratic over  $\mathbf{t}$  gives the random vector

$$\mathbf{T}_*(u) = \mathbf{I}_{\theta\theta}^{-1}(\mathbf{S}_{\theta,n} - \mathbf{i}_{\theta\kappa} u).$$

The nuisance maximizer of  $Q_n$  at  $u = 0$  is  $\mathbf{T}_*(0)$ . We define the gain over this null maximum by

$$q_n(u) = Q_n\{u, \mathbf{T}_*(u)\} - Q_n\{0, \mathbf{T}_*(0)\}.$$

Thus the corresponding log-likelihood difference is

$$\begin{aligned} \ell_n\left(\frac{u}{\sqrt{n}}, \theta_0 + \frac{\mathbf{T}_*(u)}{\sqrt{n}}\right) - \ell_n\left(0, \theta_0 + \frac{\mathbf{T}_*(0)}{\sqrt{n}}\right) \\ = q_n(u) + o_P(1). \end{aligned} \quad (\text{B.3})$$

Substituting the two maximizers into  $Q_n$  cancels the common nuisance-only term and gives

$$q_n(u) = u W_n - \frac{1}{2} j_{\text{eff}} u^2,$$

where the normalized efficient score is

$$W_n = S_{\kappa,n} - \mathbf{i}_{\theta\kappa}^\top \mathbf{I}_{\theta\theta}^{-1} \mathbf{S}_{\theta,n}.$$

This subtracts the linear projection of the score for  $\kappa$  on the nuisance scores. From (B.1),

$$W_n \xrightarrow{d} \mathcal{N}(0, j_{\text{eff}}).$$

Over  $u \in \mathbb{R}$  the quadratic would be maximized at  $W_n/j_{\text{eff}}$ , of either sign, whereas over  $u \geq 0$  the maximizer is

$$U_* = \frac{[W_n]_+}{j_{\text{eff}}},$$

which is zero whenever  $W_n \leq 0$ . Twice the maximized likelihood difference is

$$T = \frac{[W_n]_+^2}{j_{\text{eff}}} + o_{\mathbf{P}}(1),$$

so that, for a standard normal  $Z_0$ ,

$$T \xrightarrow{d} [Z_0]_+^2.$$

The right-hand side is zero when  $Z_0 \leq 0$ , an event of probability one half, and conditional on  $Z_0 > 0$  its square is chi-squared with one degree of freedom. This establishes Theorem 2.1 and its rejection rule.

### B.3 Local power

The local log-likelihood expansion (B.2) gives the power of the test, as Böhnmstedt and Gampe (2019) show for the frailty variance of the gamma-Gompertz model. Under local alternatives  $\kappa_n = u/\sqrt{n}$  with  $u > 0$  fixed, Le Cam's third lemma gives  $W_n \xrightarrow{d} \mathcal{N}(u j_{\text{eff}}, j_{\text{eff}})$  (Van der Vaart, 2000), so that  $T \xrightarrow{d} [Z_0 + u\sqrt{j_{\text{eff}}}]_+^2$ . Because  $\chi_{1,1-2\alpha}^2 = z_{1-\alpha}^2$ , the rule (8) rejects with probability tending to  $\mathbb{P}(Z_0 + u\sqrt{j_{\text{eff}}} > z_{1-\alpha}) = \Phi(u\sqrt{j_{\text{eff}}} - z_{1-\alpha})$ .

For a target power  $\pi \in (\alpha, 1)$ , inverting the preceding limiting power gives

$$u\sqrt{j_{\text{eff}}} = z_{1-\alpha} + z_{\pi}.$$

Substituting  $\kappa = u/\sqrt{n}$  gives the local detection approximation in Section 2.6. We evaluate the information at the null, so this inversion does not give an exact finite-sample threshold.

### B.4 Connection to the profile score

To connect this with the profile quantities of Section 2.1, note first that  $\boldsymbol{\theta}$  sits at its unconstrained maximum inside  $\ell_p$ , so differentiating that maximization with respect to  $\kappa$  contributes nothing and the profile score  $U$  equals the ordinary score

for  $\kappa$  evaluated at the restricted estimator  $\hat{\boldsymbol{\theta}}_{0,n}$ . Expanding its nuisance score equation gives

$$\sqrt{n}(\hat{\boldsymbol{\theta}}_{0,n} - \boldsymbol{\theta}_0) = \mathbf{I}_{\boldsymbol{\theta}\boldsymbol{\theta}}^{-1} \mathbf{S}_{\boldsymbol{\theta},n} + o_{\mathbf{P}}(1).$$

Let  $\mathbf{V}_n$  denote the observed information matrix, the negative Hessian of  $\ell_n$  at  $(0, \hat{\boldsymbol{\theta}}_{0,n})$ , with scalar entry  $v_{\kappa\kappa,n}$ , vector block  $\mathbf{v}_{\theta\kappa,n}$  and matrix block  $\mathbf{V}_{\theta\theta,n}$ . Differentiating the nuisance score equation along the profile gives

$$J = v_{\kappa\kappa,n} - \mathbf{v}_{\theta\kappa,n}^\top \mathbf{V}_{\theta\theta,n}^{-1} \mathbf{v}_{\theta\kappa,n}.$$

Combining  $\mathbf{V}_n/n \xrightarrow{\mathbf{P}} \mathbf{I}$  with the score expansion,

$$n^{-1/2}U = W_n + o_{\mathbf{P}}(1),$$

$$J/n \xrightarrow{\mathbf{P}} j_{\text{eff}}.$$

These limits justify the standardization (4) and the quadratic approximation (5), and in particular they are what permits the fitted residuals at  $\hat{\boldsymbol{\theta}}_{0,n}$ , which are not independent across ages, to be replaced by the independent residuals at  $\boldsymbol{\theta}_0$ , to which a central limit theorem applies.

## C Local information conditions

This appendix verifies the local information condition of Theorem 2.1 for the two models of Section 2 and derives the narrow-window bound used in Section 2.6.

### C.1 Gompertz–Makeham

Theorem 2.1 requires that no combination of nuisance-parameter changes can mimic a constant addition to mortality. We check that condition for the Gompertz–Makeham model and show that it holds as soon as three distinct ages are observed with positive exposure. Mimicking the constant would require an exponential, a constant and a linear function of age to be linearly dependent at three points, which Rolle's theorem excludes.

We use the nuisance parameter vector  $\boldsymbol{\theta} = (\log \lambda, \beta)^\top$ . Under  $H_0$ ,  $h_i = \lambda e^{\beta x_i}$  and

$$\mathbf{r}_i = \begin{pmatrix} h_i \\ x_i h_i \end{pmatrix}.$$

The blocks of the normalized Fisher information

matrix are

$$\begin{aligned} i_{\kappa\kappa} &= \sum_i \bar{e}_i/h_i, \\ i_{\theta\kappa} &= \begin{pmatrix} \sum_i \bar{e}_i \\ \sum_i \bar{e}_i x_i \end{pmatrix}, \\ \mathbf{I}_{\theta\theta} &= \begin{pmatrix} \sum_i \bar{e}_i h_i & \sum_i \bar{e}_i x_i h_i \\ \sum_i \bar{e}_i x_i h_i & \sum_i \bar{e}_i x_i^2 h_i \end{pmatrix}. \end{aligned}$$

A positive definite Fisher information matrix for three parameters requires at least three distinct ages with positive exposures. To see why three are enough when  $\lambda, \beta > 0$ , consider the derivative matrix

$$\mathbf{B} = \begin{pmatrix} 1 & h_1 & x_1 h_1 \\ \vdots & \vdots & \vdots \\ 1 & h_m & x_m h_m \end{pmatrix}.$$

Its columns correspond to changes in  $\kappa$ ,  $\log \lambda$  and  $\beta$ . If they were linearly dependent, constants  $\rho_1, \rho_2, \rho_3$ , not all zero, would satisfy

$$\lambda \rho_1 + \rho_2 h_i + \rho_3 x_i h_i = 0$$

at every fitted age. The factor  $\lambda > 0$  in the first coefficient only rescales that coefficient. Dividing by  $h_i = \lambda e^{\beta x_i}$  gives

$$f(x_i) = \rho_1 e^{-\beta x_i} + \rho_2 + \rho_3 x_i = 0.$$

Thus linear dependence would require the function  $f$  to have zeros at three distinct ages. By Rolle's theorem, three zeros of  $f$  require at least two zeros of  $f'$  and hence at least one zero of  $f''$  (Strang & Herman, 2016, Section 4.4). But

$$f''(x) = \rho_1 \beta^2 e^{-\beta x}$$

never equals zero if  $\rho_1 \neq 0$ . If  $\rho_1 = 0$ ,  $f$  is linear, and three zeros force  $\rho_2 = \rho_3 = 0$ . Both cases contradict the proposed dependence. The columns therefore have full rank. Positive exposure weights give a positive definite Fisher information matrix and  $j_{\text{eff}} > 0$ . With the remaining assumptions in Appendix A, Theorem 2.1 establishes Proposition 2.2.

## C.2 Information on a narrow age window

The preceding proof shows that the Makeham term is locally distinguishable from level and slope changes. It does not show how well we can distinguish them. We now show why that distinction becomes weak on a narrow age window, supplying

the bound behind Section 2.6. Write  $t = x - x_0$  and  $h_0 = \lambda e^{\beta x_0}$ . The function

$$\frac{1 + \beta x_0}{h_0} h(x) - \frac{\beta}{h_0} x h(x) = (1 - \beta t) e^{\beta t}$$

is a linear combination of  $h$  and  $xh$ . Taylor expansion gives

$$(1 - \beta t) e^{\beta t} = 1 - \frac{1}{2} \beta^2 t^2 + O(t^3).$$

This supplies a candidate in (A.4) whose residual is of order  $t^2$ . On a window with  $|t| \leq w$ , for fixed  $\beta$  and small  $w$ , it gives

$$j_{\text{eff}} \leq C w^4 \sum_i \bar{e}_i / h_i$$

for some constant  $C$  independent of  $w$ . Thus a narrow window makes the constant difficult to distinguish from level and slope adjustments when the total information weight remains bounded. This local bound does not specify the information loss for every change of age range or exposure pattern.

## C.3 Gamma-Gompertz–Makeham

We now supply the derivatives of the gamma-Gompertz hazard needed to form the Fisher information matrix when frailty is present, and then show that four distinct ages with positive exposures give that matrix full rank whenever  $\lambda\gamma \neq \beta$ . As in Appendix C.1, the argument turns on the linear independence of a small set of functions of age, and what it establishes is the local information condition rather than global identification.

We use  $\boldsymbol{\theta} = (\log \lambda, \beta, \gamma)^\top$  and write  $g$  for (11). Define the auxiliary functions

$$\begin{aligned} q(x) &= \frac{\lambda}{\beta} (e^{\beta x} - 1), \\ s(x) &= 1 + \gamma q(x), \\ q_\beta(x) &= \frac{\lambda}{\beta^2} \{ \beta x e^{\beta x} - (e^{\beta x} - 1) \}. \end{aligned}$$

Keeping the age argument explicit, differentiation gives

$$\begin{aligned} \frac{\partial g(x)}{\partial \log \lambda} &= \frac{g(x)}{s(x)}, \\ \frac{\partial g(x)}{\partial \beta} &= g(x) \left\{ x - \frac{\gamma q_\beta(x)}{s(x)} \right\}, \\ \frac{\partial g(x)}{\partial \gamma} &= -\frac{g(x) q(x)}{s(x)}. \end{aligned}$$

Substituting these into (A.2) determines the normalized Fisher information matrix. When  $\gamma_0 > 0$ ,

the frailty variance can be both increased and decreased by a sufficiently small amount, and the other nuisance parameters satisfy (A1). We exclude  $\lambda\gamma = \beta$ , because it makes  $g$  constant and violates regular identification of the added constant.

Four distinct ages with positive exposures then suffice for full rank. Write  $A = 1 - \gamma\lambda/\beta$  and  $B = \gamma\lambda/\beta$ , so that  $s(x) = A + Be^{\beta x}$ , with  $A \neq 0$  precisely when  $\lambda\gamma \neq \beta$ . Multiplying row  $i$  by the positive factor  $s(x_i)^2/(\lambda e^{\beta x_i})$  leaves the rank unchanged and turns the four derivative columns for  $(\kappa, \log \lambda, \beta, \gamma)$  into

$$\begin{aligned}\kappa &\mapsto \frac{A^2}{\lambda} e^{-\beta x} + \frac{2AB}{\lambda} + \frac{B^2}{\lambda} e^{\beta x}, \\ \log \lambda &\mapsto 1, \\ \beta &\mapsto Ax + \frac{B}{\beta} (e^{\beta x} - 1), \\ \gamma &\mapsto \frac{\lambda}{\beta} (1 - e^{\beta x}),\end{aligned}$$

where the terms in  $xe^{\beta x}$  cancel in the third column. Using rows in the order  $e^{-\beta x}, 1, x, e^{\beta x}$  and columns in the order  $\kappa, \log \lambda, \beta, \gamma$ , their coefficients form the matrix

$$\mathbf{C} = \begin{pmatrix} A^2/\lambda & 0 & 0 & 0 \\ 2AB/\lambda & 1 & -B/\beta & \lambda/\beta \\ 0 & 0 & A & 0 \\ B^2/\lambda & 0 & B/\beta & -\lambda/\beta \end{pmatrix}.$$

Expansion along its first row and then the row containing  $A$  gives

$$\det \mathbf{C} = \frac{A^2}{\lambda} A \left( -\frac{\lambda}{\beta} \right) = -\frac{A^3}{\beta}.$$

Thus the transformation between these basis functions and the derivative columns is invertible exactly when  $\lambda\gamma \neq \beta$ .

It remains to check that the four basis functions are linearly independent when evaluated at four distinct ages  $x_1 < x_2 < x_3 < x_4$ . If they were dependent, a linear combination with coefficients not all zero,

$$f(x) = \rho_1 e^{-\beta x} + \rho_2 + \rho_3 x + \rho_4 e^{\beta x},$$

would satisfy  $f(x_i) = 0$  for  $i = 1, 2, 3, 4$ . Rolle's theorem would then give at least three zeros of  $f'$  between these ages and at least two zeros of  $f''$  (Strang & Herman, 2016, Section 4.4). However,

$$f''(x) = \beta^2 (\rho_1 e^{-\beta x} + \rho_4 e^{\beta x})$$

has at most one unless  $\rho_1 = \rho_4 = 0$ , in which case  $f$  is linear and its four zeros force  $\rho_2 = \rho_3 = 0$ . The derivative matrix therefore has full rank, positive exposure weights give a positive definite normalized Fisher information matrix, and  $j_{\text{eff}} > 0$ . This covers  $\gamma_0 = 0$  as well as  $\gamma_0 > 0$ . As in Appendix C.1 the result is the local information condition, with global identification a separate requirement in (A2). When  $\gamma_0 > 0$  and the remaining assumptions hold, Theorem 2.1 establishes Proposition 2.3.

## D Distribution under $H_0$ when the true frailty variance is zero

When the true frailty variance is zero, two parameters sit on boundaries at once, changing the set of admissible local parameter displacements. Instead of maximizing a quadratic over a half-line, we maximize it over a quadrant under the alternative and over one edge of that quadrant under the null.

### D.1 Limiting statistic

At a true parameter with  $\kappa = \gamma = 0$ , we collect the constrained parameters in  $\boldsymbol{\psi} = (\kappa, \gamma)^\top$  and the nuisance parameters in  $\boldsymbol{\eta} = (\log \lambda, \beta)^\top$ . The vector  $\boldsymbol{\eta}$  can be varied in every direction by a sufficiently small amount. We assume the analogues of (A1)–(A4) with both nonnegativity constraints. Normalizing the Fisher information matrix by  $n$ , as in (A.2), and adjusting for estimation of  $\boldsymbol{\eta}$  gives

$$\mathbf{K} = \mathbf{I}_{\boldsymbol{\psi}\boldsymbol{\psi}} - \mathbf{I}_{\boldsymbol{\psi}\boldsymbol{\eta}} \mathbf{I}_{\boldsymbol{\eta}\boldsymbol{\eta}}^{-1} \mathbf{I}_{\boldsymbol{\eta}\boldsymbol{\psi}}, \quad (\text{D.1})$$

which we assume positive definite. At  $\kappa = \gamma = 0$ , define the derivative vector in the order  $(\kappa, \gamma, \log \lambda, \beta)^\top$  by

$$\mathbf{d}_i = (1, -h_i q_i, h_i, x_i h_i)^\top,$$

where  $q_i = q(x_i)$  as in Appendix C.3. For this parameter order, (A.2) becomes

$$\mathbf{I} = \sum_i \frac{\bar{e}_i}{h_i} \mathbf{d}_i \mathbf{d}_i^\top.$$

We partition this matrix into the first two and last two coordinates, then substitute its blocks into (D.1) to obtain  $\mathbf{K}$ .

Applying the score limit (B.1) to this four-parameter vector gives  $\mathbf{S} \sim \mathcal{N}_4(\mathbf{0}, \mathbf{I})$ . Write  $\mathbf{S} = (\mathbf{S}_{\boldsymbol{\psi}}^\top, \mathbf{S}_{\boldsymbol{\eta}}^\top)^\top$ , so  $\mathbf{S}_{\boldsymbol{\psi}}$  contains the scores for  $\kappa$  and  $\gamma$ , and  $\mathbf{S}_{\boldsymbol{\eta}}$  contains the nuisance scores for  $\log \lambda$

and  $\beta$ . The best linear prediction of  $\mathbf{S}_\psi$  from  $\mathbf{S}_\eta$  is  $\mathbf{I}_{\psi\eta}\mathbf{I}_{\eta\eta}^{-1}\mathbf{S}_\eta$ , using their covariance blocks. Subtracting this prediction gives

$$\mathbf{W} = \mathbf{S}_\psi - \mathbf{I}_{\psi\eta}\mathbf{I}_{\eta\eta}^{-1}\mathbf{S}_\eta,$$

so that

$$\mathbf{W} \sim \mathcal{N}_2(\mathbf{0}, \mathbf{K}).$$

Let  $\mathbf{z} = (z_1, z_2)^\top$  denote the fixed local displacements of  $\psi$ , so  $\kappa = z_1/\sqrt{n}$  and  $\gamma = z_2/\sqrt{n}$ . After maximizing over the local displacements of  $\eta$ , we obtain the quadratic

$$q_{\mathbf{W}}(\mathbf{z}) = \mathbf{z}^\top \mathbf{W} - \frac{1}{2} \mathbf{z}^\top \mathbf{K} \mathbf{z}.$$

Under the full model we maximize over  $\mathbf{z} \in \mathbb{R}_+^2$ , and under  $H_0$  we set  $z_1 = 0$  and maximize over  $z_2 \geq 0$ . Using the expansion (B.2) with both constraints,

$$T \xrightarrow{d} T_0(\mathbf{W}),$$

where

$$T_0(\mathbf{W}) = 2 \left\{ \sup_{\mathbf{z} \in \mathbb{R}_+^2} q_{\mathbf{W}}(\mathbf{z}) - \sup_{t \geq 0} q_{\mathbf{W}}((0, t)^\top) \right\}.$$

In the first supremum both local parameters are free to be nonnegative, while in the second the local value of  $\kappa$  is fixed at zero and the local value of  $\gamma$  is estimated. This is the quadrant-versus-edge geometry of the nuisance-boundary cases in Self and Liang (1987), and because the null is an edge rather than the origin, the difference of the two suprema is not in general distributed as a mixture of chi-squared variables.

## D.2 Explicit form and critical values

The constrained Gaussian quadratic follows from the boundary likelihood theory of Self and Liang (1987, Theorem 3). We evaluate it here for the quadrant and its null edge by considering the origin, the two edges and the interior. Writing  $k_{ij}$  for the entries of  $\mathbf{K}$ , the candidate values for twice the full maximum are

$$\begin{aligned} A_1 &= [W_1]_+^2 / k_{11}, \\ A_2 &= [W_2]_+^2 / k_{22}, \end{aligned}$$

corresponding to the two edges of the quadrant, together with

$$A_3 = \mathbf{W}^\top \mathbf{K}^{-1} \mathbf{W}$$

when  $\mathbf{K}^{-1}\mathbf{W} \in \mathbb{R}_+^2$ , corresponding to an admissible unconstrained maximizer of the strictly concave quadratic. The origin needs no separate candidate, since  $A_1$  and  $A_2$  are already nonnegative

and equal zero exactly when the corresponding edge maximum is at the origin. Collecting the admissible candidates in  $\mathcal{A}(\mathbf{W})$ ,

$$T_0(\mathbf{W}) = \max \mathcal{A}(\mathbf{W}) - A_2, \quad (\text{D.2})$$

where subtracting  $A_2$  removes the maximized contribution under  $H_0$ .

Equation (D.2) is what makes the rejection rule (12) usable. Fit the Gompertz model with  $\kappa$  and  $\gamma$  both fixed at zero, form  $\widehat{\mathbf{K}}$  from the resulting  $\widehat{\lambda}, \widehat{\beta}$  using (A.2) and the partition in (D.1), draw  $B$  replicates  $\mathbf{W} \sim \mathcal{N}_2(\mathbf{0}, \widehat{\mathbf{K}})$ , evaluate  $T_0(\mathbf{W})$  at each draw, and take the empirical  $1 - \alpha$  quantile as the critical value. Two approximations enter. The limit is asymptotic, and  $\widehat{\mathbf{K}}$  replaces  $\mathbf{K}$ , which needs the Gompertz fit to be consistent under  $\kappa = \gamma = 0$ . The  $B$  draws add Monte Carlo error to the quantile, which can be made small by taking  $B$  large.

## D.3 The role of $k_{12}$

**Uncorrelated adjusted scores.** If  $\mathbf{K}$  is diagonal, the quadratic separates into contributions from  $\kappa$  and  $\gamma$ . The maximized contribution from  $\gamma$  is then the same under both hypotheses and cancels in the likelihood ratio, leaving

$$T_0(\mathbf{W}) = [W_1]_+^2 / k_{11},$$

so the limit reverts to  $\frac{1}{2}\delta_0 + \frac{1}{2}\chi_1^2$ . Otherwise the fits of the two constrained parameters interact, the cancellation fails, and the distribution must be determined from (D.2). In particular, the critical value 2.7055, which is the 95th percentile of  $\frac{1}{2}\delta_0 + \frac{1}{2}\chi_1^2$ , cannot be assumed to be the 95th percentile of  $F_{\mathbf{K}}$ .

## Negatively correlated adjusted scores.

When  $k_{12} < 0$ , we can establish the direction of the error. We first derive the statistic for each score configuration, then calculate its mass at zero and compare its upper tail with the half-and-half law. Write  $\rho = k_{12}/\sqrt{k_{11}k_{22}}$ , and let

$$\begin{aligned} W_{1.2} &= W_1 - \frac{k_{12}}{k_{22}} W_2, \\ k_{11.2} &= k_{11} - \frac{k_{12}^2}{k_{22}}, \end{aligned}$$

with  $W_{2.1}$  and  $k_{22.1}$  defined by exchanging the indices. Then  $W_{1.2} \sim \mathcal{N}(0, k_{11.2})$  is independent

of  $W_2$ , and

$$\begin{aligned}\mathbf{W}^\top \mathbf{K}^{-1} \mathbf{W} &= \frac{W_2^2}{k_{22}} + \frac{W_{1.2}^2}{k_{11.2}}, \\ \mathbf{W}^\top \mathbf{K}^{-1} \mathbf{W} &= \frac{W_1^2}{k_{11}} + \frac{W_{2.1}^2}{k_{22.1}}.\end{aligned}$$

The components of  $\mathbf{z} = \mathbf{K}^{-1} \mathbf{W}$  are  $z_1 = W_{1.2}/k_{11.2}$  and  $z_2 = W_{2.1}/k_{22.1}$ , and they satisfy  $k_{12}z_1 + k_{22}z_2 = W_2$ .

**Case 1:**  $W_2 > 0$ . If  $W_{1.2} > 0$ , the unconstrained maximizer has  $z_1 > 0$ . Since  $k_{12} < 0$ , its other coordinate also satisfies

$$z_2 = \frac{W_2 - k_{12}z_1}{k_{22}} > 0.$$

Both coordinates are admissible, so the full maximum is  $A_3$ . Subtracting the null maximum gives

$$T_0 = A_3 - A_2 = \frac{W_{1.2}^2}{k_{11.2}}.$$

If instead  $W_{1.2} \leq 0$ , then  $W_1 \leq (k_{12}/k_{22})W_2 < 0$ . The first edge gives  $A_1 = 0$ , and the interior cannot improve on the null edge. Hence  $T_0 = 0$ .

**Case 2:**  $W_2 \leq 0$ . Here the null maximum is  $A_2 = 0$ . If  $W_1 \leq 0$  as well, both linear terms in the quadratic are nonpositive on the quadrant. The full maximum is therefore at the origin, and  $T_0 = 0$ .

If  $W_1 > 0$ , the first edge gives  $A_1 = W_1^2/k_{11}$ . The interior maximum is admissible exactly when  $W_{2.1} \geq 0$ . To check the other coordinate in that event, observe that

$$k_{22}W_1 - k_{12}W_2 \geq \frac{W_1 \det \mathbf{K}}{k_{11}} > 0.$$

Thus  $z_1 > 0$  too. The second quadratic decomposition then adds  $W_{2.1}^2/k_{22.1}$  to  $A_1$ ; otherwise the first edge remains the maximum. Combining the cases gives

$$T_0 = \begin{cases} [W_{1.2}]_+^2/k_{11.2}, & W_2 > 0, \\ W_1^2/k_{11} + [W_{2.1}]_+^2/k_{22.1}, & W_2 \leq 0, W_1 > 0, \\ 0, & W_2 \leq 0, W_1 \leq 0. \end{cases}$$

**Probability mass at zero.** Sheppard's formula for the orthant probabilities of a bivariate normal distribution gives  $\mathbb{P}(W_1 \leq 0, W_2 \leq 0) = \frac{1}{4} + \arcsin(\rho)/(2\pi)$  for any correlation  $\rho$  (Sheppard, 1899; Tong, 2012). Together with  $\mathbb{P}(W_2 > 0, W_{1.2} \leq 0) = \frac{1}{4}$ , which holds because  $W_2$  and

$W_{1.2}$  are independent and centred, the piecewise form gives

$$\mathbb{P}(T_0 = 0) = \frac{1}{2} + \frac{\arcsin \rho}{2\pi} < \frac{1}{2}.$$

**Upper-tail comparison.** For the upper tail, fix  $c > 0$ , let  $V \sim \chi_1^2$  and  $a = \sqrt{ck_{11}}$ , so that  $\frac{1}{2}\mathbb{P}(V > c) = \mathbb{P}(W_1 > a)$ . By the independence of  $W_{1.2}$  and  $W_2$ , the first case contributes  $\mathbb{P}(T_0 > c, W_2 > 0) = \frac{1}{4}\mathbb{P}(V > c) = \frac{1}{2}\mathbb{P}(W_1 > a)$ . In the second case  $T_0 \geq W_1^2/k_{11}$ , so it contributes at least  $\mathbb{P}(W_2 \leq 0, W_1 > a)$ . Therefore

$$\begin{aligned}\mathbb{P}(T_0 > c) - \frac{1}{2}\mathbb{P}(V > c) &\geq \mathbb{P}(W_1 > a) \\ &\quad \times \left\{ \mathbb{P}(W_2 \leq 0 \mid W_1 > a) - \frac{1}{2} \right\} \\ &> 0.\end{aligned}$$

because, given  $W_1 = w > 0$ ,  $W_2$  is normal with mean  $(k_{12}/k_{11})w < 0$ . Taking  $c = \chi_{1,1-2\alpha}^2$  shows that the rule (8) has asymptotic rejection probability above  $\alpha$  at every level. When  $k_{12} = 0$  the bound holds with equality, in agreement with the diagonal case above.

#### D.4 Frailty variances near zero

For frailty variances close to but not equal to zero, set  $\kappa = 0$  and  $\gamma_n = \tau/\sqrt{n}$  with fixed  $\tau \geq 0$ . Expanding the log-likelihood about  $\kappa = \gamma = 0$ , the limiting adjusted score is

$$\mathbf{W}_\tau \sim \mathcal{N}_2 \left( \mathbf{K}(0, \tau)^\top, \mathbf{K} \right),$$

and the statistic satisfies

$$T \xrightarrow{d} T_0(\mathbf{W}_\tau).$$

The limiting distribution can therefore depend on  $\tau$ , which describes how the true frailty variance approaches zero as  $n \rightarrow \infty$  under (A.1). The result at exactly  $\gamma = 0$  and the result at a fixed  $\gamma > 0$  do not by themselves bound the rejection probability across all such sequences. Controlling the type I error when  $\gamma$  is unknown requires an argument that applies uniformly across these values, and selecting a reference distribution according to whether  $\hat{\gamma} = 0$  does not supply one. The results remain valid when frailty is fixed at zero by specification, or when its true variance is fixed and positive.

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